CprE 458/558: Real-Time Systems

Overload Handling in Real-Time Systems – Part 2

Feedback Control based EDF Scheduling

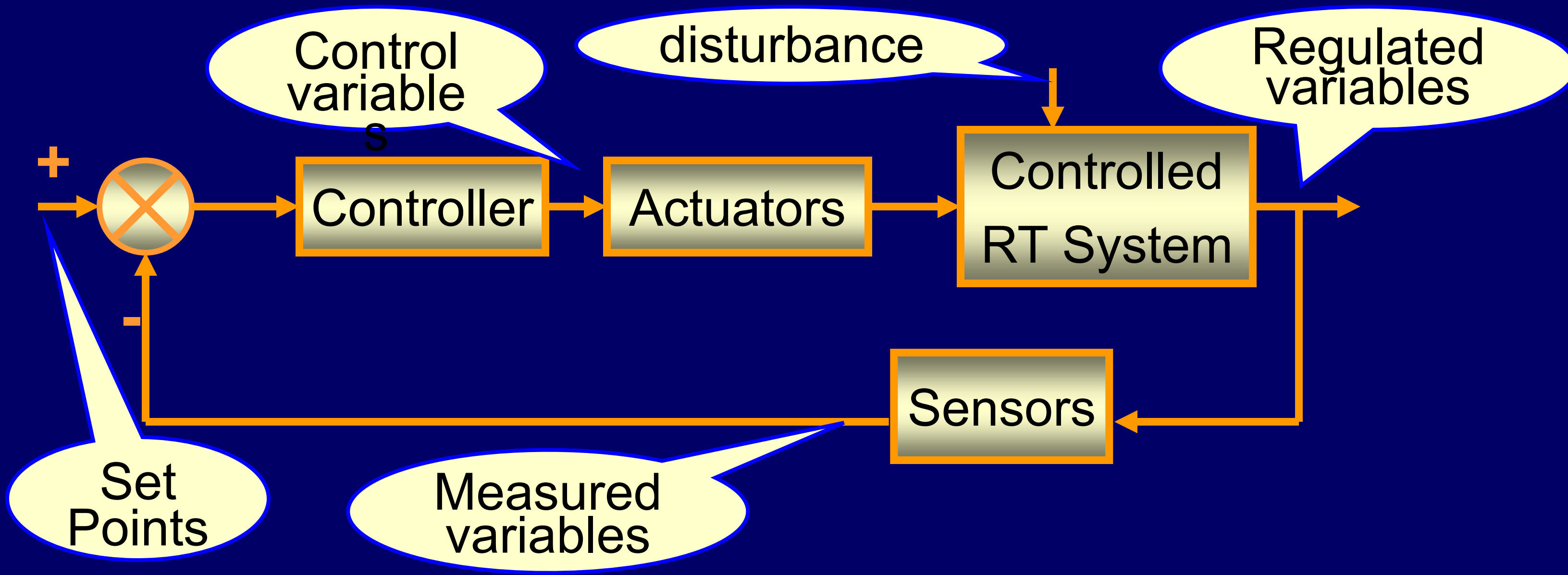
Feedback scheduling – motivation

One of the very successful areas in addressing performance in the presence of uncertainty (e.g., workload or fault) is that of Robust Control. <u>Feedback of measured quantities to correct the behavior of a system has been a powerful concept</u> that has made technological advances in applications such as amplifiers and avionics.

Through concerted use of feedback control and its theoretical development, the concept has been used to deal with uncertainty inherent in most systems.

It also needs to be stated that if a system characteristics is known precisely, then the feedback strategies are not useful; the open-loop strategies will outperform their feedback counterpart.

Feedback control technique



Feedback System Operation

- A typical feedback control system is composed of a controller, a plant to be controlled, actuators, and sensors.
- Controlled/regulated variable, the quantity of the output that is measured and controlled/regulated.
- The set point represents the correct value of the controlled variable.
- The difference between the current value of the controlled variable and the set point is the error.
- The manipulated/control variable is the quantity that is varied by the controller so as to affect the value of the controlled/regulated variable.

Feedback system operation (contd.)

The system is composed of a feedback loop as follows.

- The system periodically measures and compares the controlled variable to the set point to determine the error.
- The controller computes the required control with the control function of the system based on the error.
- The actuators change the value of the manipulated variable to control the system.

FC-EDF

- Task Model

$$T_i = (I, ET, A, S, D)$$

- Each Ti has logical versions: $I = (T_{i1}, T_{i2}, \ldots T_{ik})$

- Each version has different execution time:

$$ET = \{ET_{i1}, ET_{i2}, \ldots ET_{ik}\}$$

suppose $ET_{i1} \geq ET_{i2} \geq \ldots \geq ET_{ik}$

- Each version has different accuracy: $A = \{A_{i1}, A_{i2}, \ldots A_{ik}\}$

- Each task has a soft deadline D_i and a start time S_i

- Different versions of a task are called <u>service levels</u>.

- A version with longer execution time and better accuracy is called a higher service level.

FC-EDF -- Variables

- Set point: desired miss ratio

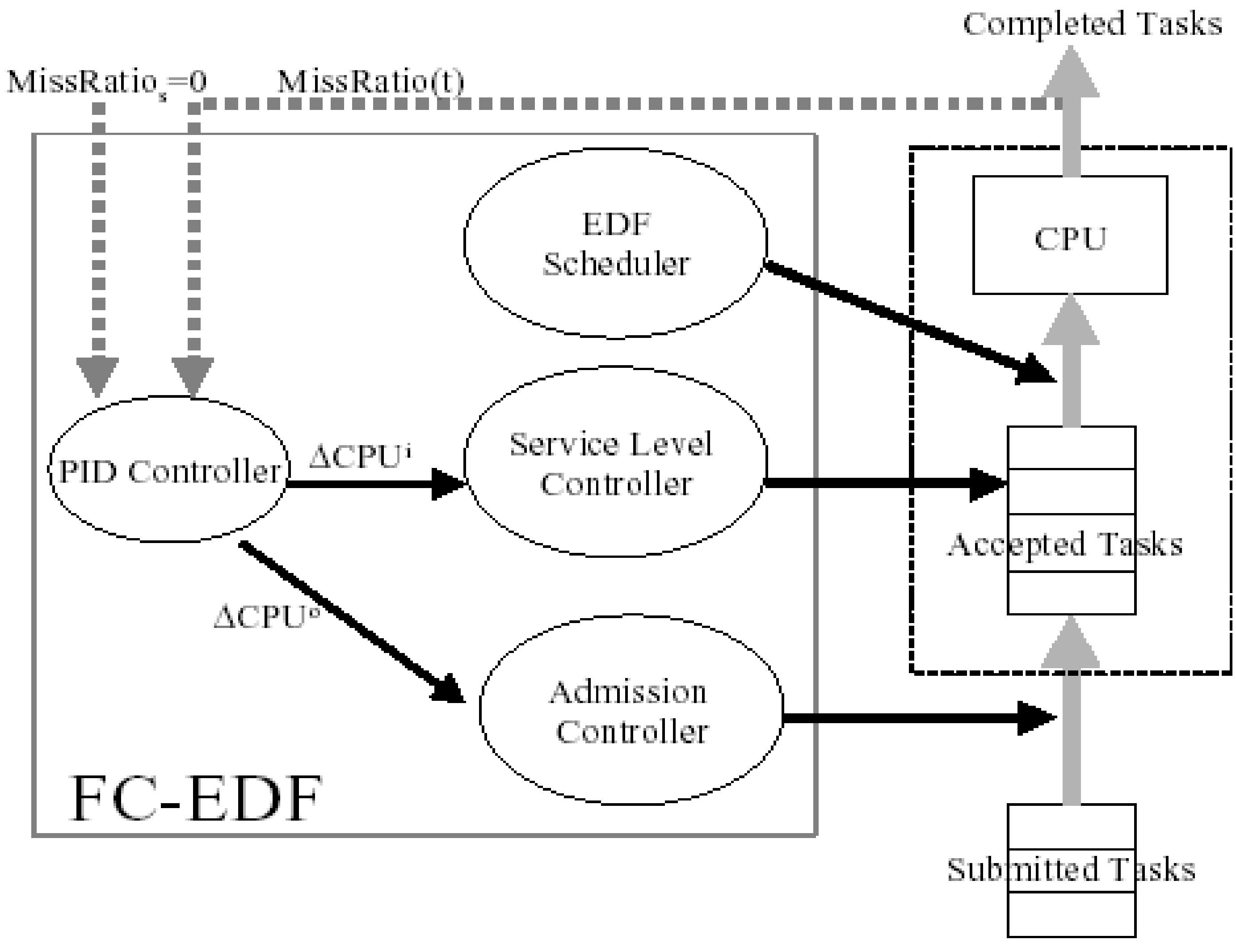
$$\left(miss\ ratio = \frac{\#\, missed\ tasks}{\#\, submitted\ tasks} \right)$$

- Regulated/Measured variable: miss ratio
- Control variable: requested CPU utilization

$$requested\ CPU\ utilization = \frac{execution\ time}{deadline - current\ time}$$

- Actuators: Server Level Controller & Admission Controller– use server level controller, if the requirements are not satisfied, use admission controller.

FC-EDF Schematic



FC-EDF

- PID Parameters Tuning
  - Simulation experiments
  - Modeling analysis
  - Adaptive control to tune the parameters on-line

[Read the paper for details]

Reference: C. Lu, J.A. Stankovic, G. Tao, and S.H. Son, "Design and Evaluation of a Feedback Control EDF Scheduling Algorithm," In Proc. Real-Time Systems Symp. pp.56-67, 1999.

Case study:

Feedback-based (m,k)-RMS Scheduler [3]

Task model and performance index

- Task Model - firm periodic tasks [1,2]

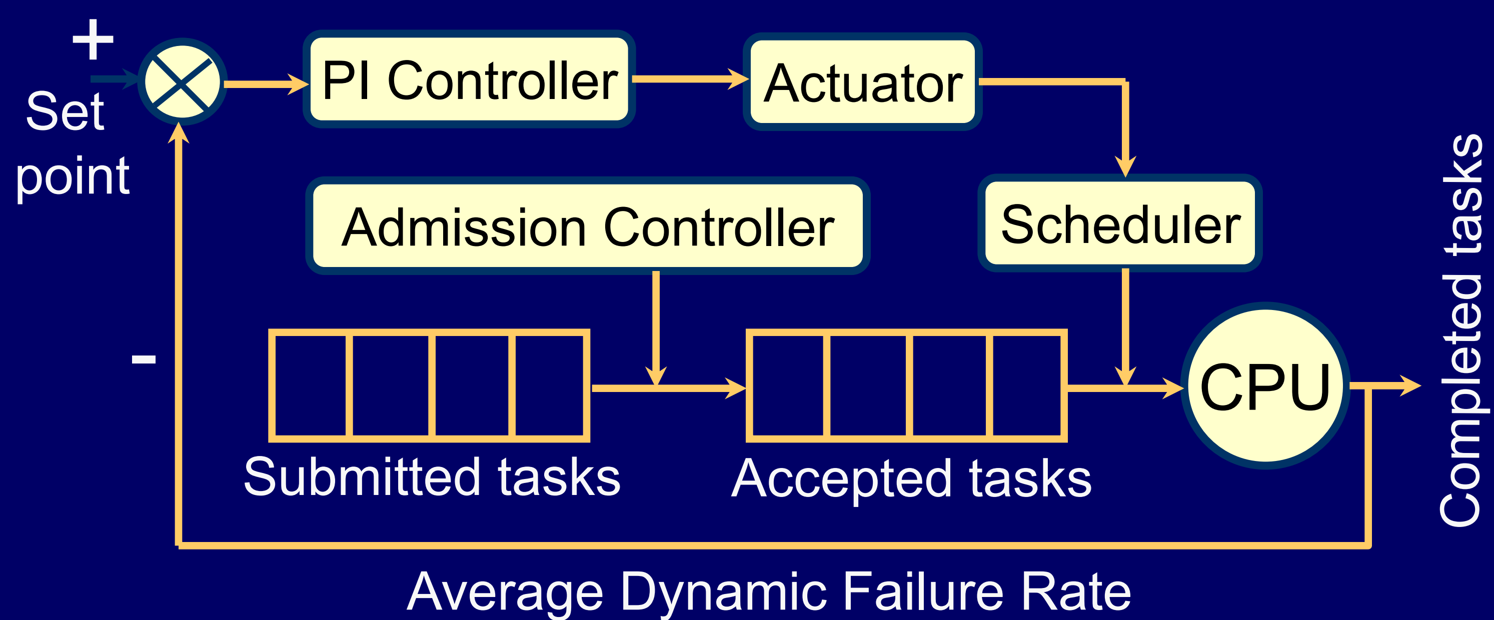
$$T_i = < c_i, p_i, m_i, k_i > \qquad c_i : comp.\,time \qquad p_i : period$$

- Tasks should meet m_i deadlines for every k_i consecutive instances
- Performance Index
  - **Dynamic Failure Rate (DFR):** for a task Ti, it is the percentage of instances of the task miss their (m,k) guarantee.
  - **Marginal Quality Received (MQR):**

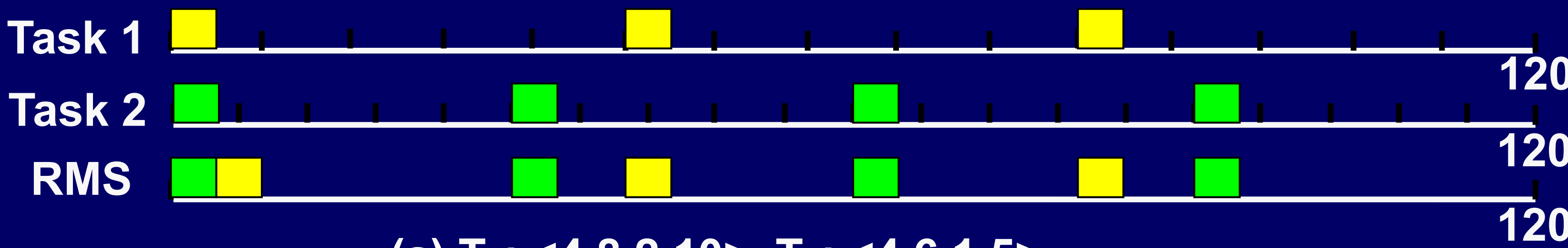
$$MQR_i = \frac{\overline{m_i} - m_i}{k_i - m_i} \qquad \overline{m_i} : the\ actual\ value\ used \qquad MQR_i : MQR\ of\ task\ T_i$$

    - To maximize the quality of tasks during overloading, $\overline{m_i}$ is increased as much as possible

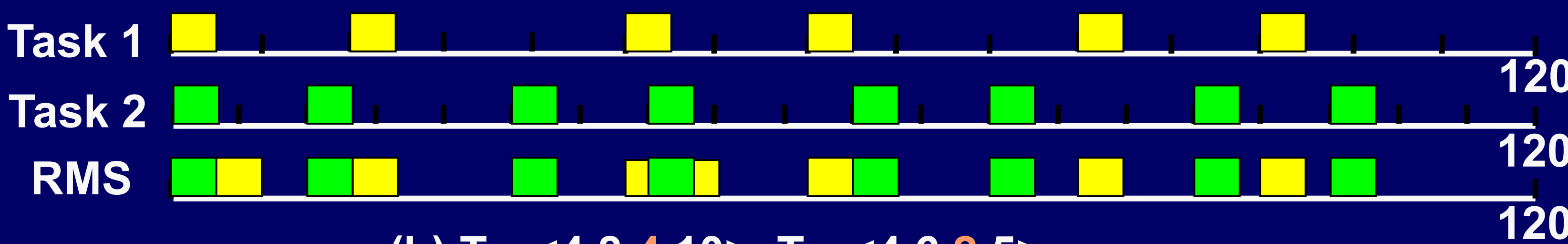
Feedback-based Adaptive Scheduler Architecture



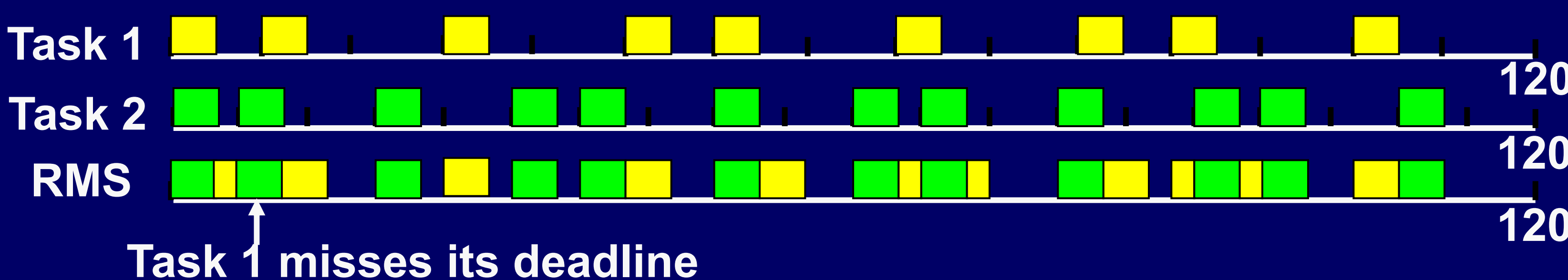
Scheduling Example



(a) T_1: <4,8,2,10> T_2: <4,6,1,5>



(b) T_1: <4,8,4,10> T_2: <4,6,2,5>



(c) T_1: <4,8,6,10> T_2: <4,6,3,5>

Example (Cont.)

- We can increase the $\overline{m_i}$ values to increase the QoS when the system is underloaded, and decrease the $\overline{m_i}$ values to handle overloading.
- Feedback method can be used to adjust the $\overline{m_i}$ values.
  - Regulated/measured variable: DFR
  - Set point: desired value of DFR
  - Control variable: estimation factor, etf, of $\overline{m_i}$

Feedback-based Adaptive scheduling (Cont.)

- Admit tasks based on minimum quality requirement
- The actual execution time of tasks are normally less than or equal to the worst case execution time used in the admission test
  - Try to increase the quality as much as possible
  - Use feedback method to adjust $\overline{m_i}$
- Non-zero set point is used – achieve high CPU utilization and low dynamic failure rate
- DFR is zero with respect to m_i – DFR is changed with respect to the current $\overline{m_i}$ later

Feedback control algorithm

- $\Delta etf_t = K(DFR_s - DFR_{t-1})$
- $etf_t = etf_{t-1} + \Delta etf_t$
- $\overline{m_{it}} = m_i + etf_t(k_i - m_i)$

$etf : estimation\ \ factor$

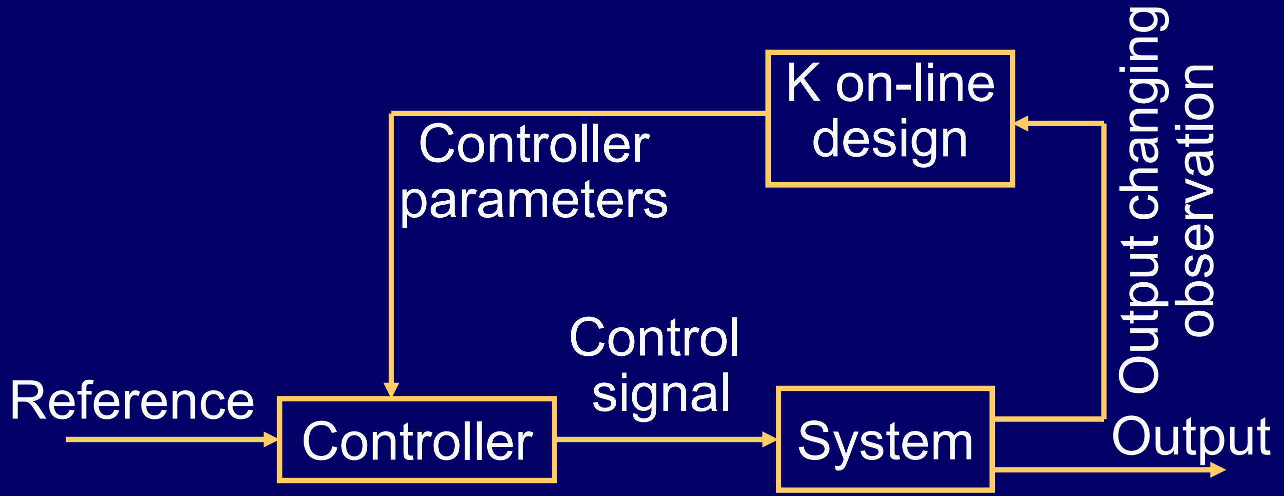
$K : controller.parameter$

$confine\, etf_t\ in\, [0,1]$

$etf_{inital} = 0.0$

Online controller design

- Initial Value of K: $K = \frac{etf_{\max}}{DFR_s - 0.0} = \frac{1.0}{0.1 - 0.0} = 10$
- Halve K when DFR fluctuate across set point

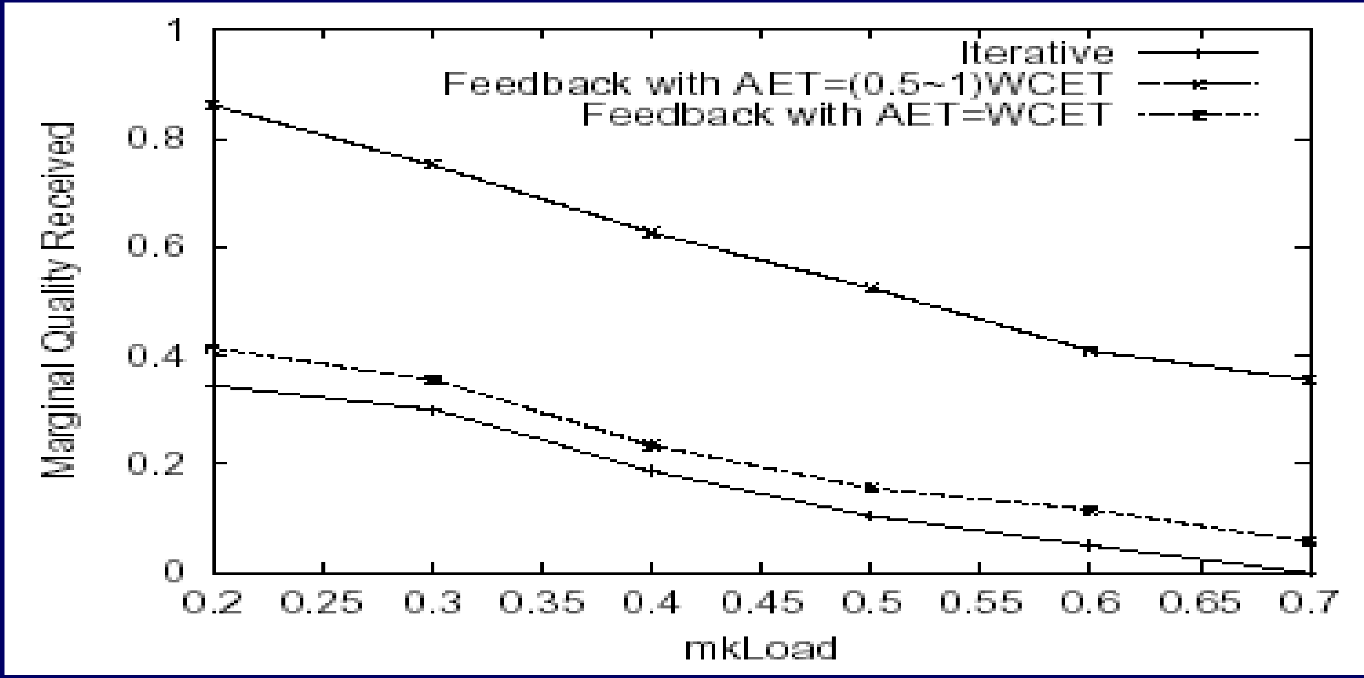


- K
  - high value will lead to fluctuation
  - Low value will lead to a long time to reach the final value

Simulation studies

$$Load = \sum_{i=1}^{n} \frac{c_i}{p_i} \qquad mkLoad = \sum_{i=1}^{n} \frac{c_i \cdot m_i}{p_i \cdot k_i}$$

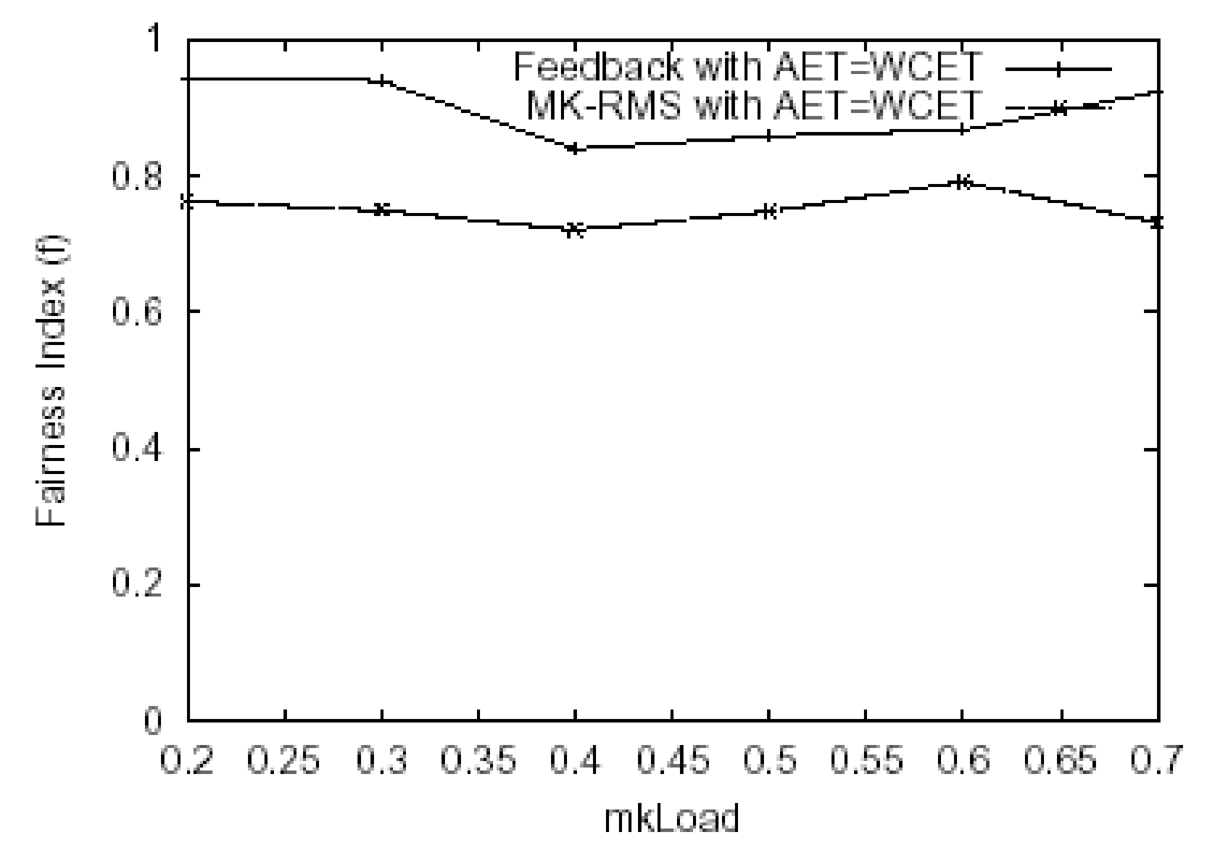
- Feedback algorithm vs. iterative algorithm



MQR performance: Load = 1.1 and MKLoad varied

- MQR decreases as MKLoad increases
- ACET < WCET can be exploited to increase MQR
- Feedback algo offers better MQR than non-feedback algo

Simulation studies (Cont.)



Fairness (f):

– Fairness obtained by the feedback approach is higher than that obtained by non-feedback algo (MK-RMS)

Overload handling in RT Systems -- Summary

- Imprecise computation & (m,k)-firm model
  - Offers scheduling flexibility to achieve graceful degradation (i.e., means to achieve predictable timing faults without violating system spec)
  - Applicable only to a class of applications
- Feedback-based resource management
  - Periodically monitors system performance
  - Adapts/controls system operation
  - Suitable sensing, control, and regulated variables needed

References

[1] C. Lu, J.A. Stankovic, G. Tao, and S.H. Son, "Design and Evaluation of a Feedback Control EDF Scheduling Algorithm," In Proc. Real-Time Systems Symp. pp.56-67, 1999.

[2] Overload management in real-time control applications using (m, k)-firm guarantee Ramanathan, P.; IEEE Transactions on Parallel and Distributed Systems, Volume 10, Issue 6, June 1999 Page(s):549 – 559.

[3] Suzhen Lin, Ph.D Dissertation, ISU, 2005.